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Runtime Validity Estimation for the Integration of Hybrid Learned Models into Vehicle Dynamics Control

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Abstract

Contents

1 Introduction **4**

1.1 Context and motivation 5
1.2 Research problem 7
1.3 Scope of the problem (Priority 1) 7
1.4 Research gap 7
1.5 Research question 7
1.6 Specific research questions 7
1.7 Research hypothesis 8

2 Objectives **9**

2.1 Overall objective 9
2.2 Specific objectives 9

3 Fundamentals of vehicle dynamics and control **11**

3.1 Vehicle dynamics relevant to control 11
3.2 Control-oriented models and model mismatch 11
3.3 Hybrid models: physics and learned residuals 11
3.4 Model predictive control 11
3.5 Control barrier functions as an optional extension 11

4 Learned information and runtime validity **12**

4.1 Learned information relevant to the controller 12
4.2 Predictive uncertainty 12
4.2.1 Aleatoric uncertainty 12
4.2.2 Epistemic uncertainty 12
4.2.3 Calibration 12
4.3 OOD and distribution shift in vehicle dynamics 12
4.3.1 Unrepresented physical regime 12
4.3.2 Dynamics OOD 12
4.3.3 Input OOD 12
4.3.4 Sensor failure or degradation 12

4.4 Runtime validity and confidence . . . 12
4.5 Control-relevant validity . . . 12

5 State of the art **13**
5.1 Literature review strategy . . . 13
5.2 Learned and hybrid models for vehicle dynamics control . . . 13
5.3 Uncertainty-aware learned dynamics for control . . . 13
5.4 OOD, distribution shift, and runtime monitoring . . . 13
5.5 Validity-aware and confidence-aware control . . . 13
5.6 Critical synthesis and research gap . . . 13

6 Conceptual framework and proposed architecture **14**
6.1 The shift–validity–control chain . . . 14
6.2 Hybrid prediction model . . . 14
6.3 Validity estimator . . . 14
6.4 Integration with MPC/NMPC . . . 14
6.5 Optional extensions . . . 14
6.5.1 Modulation of a CBF . . . 14
6.5.2 Data acquisition and retraining . . . 14

7 Research and validation methodology (Priority 1) **15**
7.1 Overall strategy . . . 15
7.2 Distribution shift scenarios . . . 16
7.2.1 Unknown change in friction . . . 16
7.2.2 Extension of the speed range . . . 16
7.2.3 Sensor failure or degradation . . . 16
7.2.4 Input OOD . . . 16
7.3 Learned model and uncertainty . . . 16
7.4 Construction and evaluation of the validity signal . . . 16
7.5 Validity-aware integration with MPC/NMPC . . . 16
7.6 Baselines and ablation study . . . 16
7.7 Evaluation metrics . . . 16
7.7.1 Model metrics . . . 16
7.7.2 Uncertainty, calibration, and OOD metrics . . . 16
7.7.3 Closed-loop metrics . . . 16
7.8 Empirical robustness criterion . . . 16
7.9 Level of assurance and scope of the conclusions . . . 17
7.10 Statistical analysis and reproducibility . . . 17

8 Preliminary results **18**
8.1 Feasibility of learning vehicle dynamics models . . . 18
8.2 Uncertainty and OOD behavior under distribution shifts . . . 18
8.3 Preliminary integration with control . . . 18
8.4 Discussion of preliminary results . . . 18

9 Expected contributions **19**
9.1 Control-relevant validity estimation . . . 19
9.2 Validity-aware vehicle control . . . 19
9.3 Runtime validity characterization . . . 19
9.4 Experimental taxonomy of shifts in vehicle dynamics . . . 19
9.5 Potential extensions . . . 19

10 Work plan (Priority 1) **20**
10.1 Remaining activities . . . 20
10.2 Schedule . . . 20
10.3 Decision milestones . . . 20

11 Risks, limitations, and mitigation strategies **21**
11.1 Risk of redundancy between uncertainty and OOD . . . 21
11.2 Risk of poor calibration or weak correlation with error . . . 21
11.3 Risk that horizon adaptation does not improve closed-loop performance . . . 21
11.4 Computational risk . . . 21
11.5 Risk of scope expansion . . . 21
11.6 Limitations of formal guarantees . . . 21

12 Conclusions of the qualifying examination **22**

Chapter 1

Introduction

Consider a driver traveling along a highway at 100 km/h when it starts to rain. As visibility decreases, the driver decides to pull over onto the shoulder. During the maneuver, however, water on the road may reduce tire–road grip, making it harder to change course. What began as an attempt to stop the journey can, under these conditions, lead to a loss of control and a risk of collision with other vehicles or roadside barriers.

The difficulty in this example lies in carrying out a maneuver when the available grip changes. Steering and braking rely on forces transmitted through tire–road contact, so a change in those conditions can alter the vehicle's ability to follow the intended path [1].

Available grip varies with the road surface and the conditions encountered along the route. Studies of pavement friction link skid resistance to surface texture and road characteristics [19, 10]. Water on the pavement can impair tire–road contact, and ruts can retain water and increase the risk of hydroplaning [15].

These limits matter when a driver needs to change lanes at short notice or swerve to avoid an obstacle. The lateral forces required to follow the intended path may exceed the available tire–road force capacity [1]. At high speed and low grip, this limitation can lead to lateral sliding and yaw instability, compromising the maneuver [9].

Vehicle control systems assist the driver with braking and maintaining stability. The anti-lock braking system (ABS) aims to prevent wheel lock during braking [4]. Electronic stability control (ESC) helps correct deviations between vehicle motion and the driver's intended direction; Bosch's Electronic Stability Program (ESP), introduced in 1995, is one implementation of this function [3]. Development has since extended to coordinating braking, steering, and propulsion. This coordination offers more ways to influence vehicle motion and requires information about its response, including states and parameters that may need to be estimated during operation [16].

1.1 Context and motivation

The vehicle's response also depends on its mass, loading, and tire properties. Friction and axle loads affect braking-distance prediction [23], while mass and tire cornering stiffness enter the description of lateral and yaw motion [22]. In the linear range, cornering stiffness is the slope relating lateral tire force to slip angle [16]. It therefore describes a different aspect of tire behavior from the friction limit: the latter bounds the available forces, while stiffness influences how lateral force develops with slip angle. Changes in these quantities can affect both the forces available for a maneuver and the motion that results from them.

Studies of adaptive vehicle control illustrate the consequences of these variations. Xu et al. evaluate trajectory tracking in simulations on roads with low and changing friction, reporting lower lateral and heading errors with their adaptive strategy than with the comparison controllers [20]. Yeneneh et al. use online parameter estimation to update a lateral-tracking model that includes mass and cornering stiffness. In their simulations, the adaptive controller reduces tracking errors relative to conventional predictive control and linear quadratic regulation [22]. These results motivate considering changes in both the road environment and the vehicle itself, and examining whether the description of dynamics used for control remains representative.

Predictive controllers use a model to relate control inputs to the vehicle response they expect. Model predictive control (MPC), including its nonlinear formulation (NMPC), uses those predictions to choose inputs over a finite horizon while accounting for constraints [17]. The model must provide enough detail to support these decisions while keeping the control problem computationally tractable. Chen et al., for example, use a simplified bicycle model to balance prediction accuracy and computational cost [7]. Such simplifications can introduce discrepancies between predicted and actual motion. Pan et al. identify model structure and parameter errors as sources of this model mismatch and account for them in a robust predictive controller [14].

Uncertainty about a parameter's value need not imply that the parameter is changing. A discrepancy between its actual and assumed values can affect predictions even when its physical value remains constant. Yeneneh and Sufe consider uncertainty in mass, front and rear tire cornering stiffness, and road friction within a robust predictive-control formulation [21]. Li et al. address uncertainty in cornering stiffness and yaw moment of inertia in a controller with longitudinal and lateral coupling, evaluated through hardware-in-the-loop tests [13]. These studies motivate accounting for parameter uncertainty when using a simplified model for control.

It is therefore useful to distinguish the vehicle's physical parameters, the nominal values used by the model, and estimates obtained from measurements during operation. Estimation can reduce parameter error, but it does not remove the model's structural approximations. Chen et al. reflect this distinction by combining calibration of a nominal

physical model with compensation for remaining unmodeled dynamics [6]. Their formulation illustrates why improving prediction may require addressing discrepancies that parameter calibration alone does not capture.

Hybrid modeling addresses part of this mismatch by combining physical equations with information learned from data [8, 24]. In a model with an additive learned residual, the physical component provides a nominal prediction, and the learned component estimates a correction from the difference between that prediction and the observed response [11, 7]. The correction can represent effects that the simplified model does not describe adequately. In Kabzan et al.'s formulation, its training targets are obtained from measured state transitions and nominal predictions, allowing the model error to be learned without identifying its physical causes separately [11].

Kabzan et al. use Gaussian process regression for this correction in autonomous racing. During tests on a full-size race car, the corrected model maintained an approximately constant average prediction error as driving became more aggressive over successive laps, while the error of the nominal model increased [11]. Learning can also be assigned to individual force components. Zhou et al. combine a neural network model of traction with physical longitudinal dynamics and an estimator for vehicle mass and a modified rolling friction coefficient. Their evaluation includes load and road conditions outside those used for training [24]. This is a hybrid force-modeling and parameter-estimation architecture, rather than an additive residual formulation.

Other hybrid approaches fuse physical and neural models for dynamic prediction or couple a neural estimator with a physical observer for sideslip estimation [8, 2]. Physical knowledge can also guide training: Wang et al. include residuals of force-equilibrium equations in the learning objective of a vehicle dynamics model [18]. These uses of learning serve different roles. In this thesis, the focus is an additive residual within the controller's prediction model, evaluated through both prediction accuracy and its effect on control.

Consider, as a conceptual example, a residual trained with a particular vehicle load and later used on a journey with additional passengers or cargo. If the nominal model and residual remain unchanged, the learned correction may no longer describe the discrepancy under the new load. Changes in tire properties can raise a similar difficulty: familiar speeds and steering inputs need not produce the same response when the underlying parameters differ [21].

Loading may differ between journeys, and tire characteristics may evolve during driving. Kalaria et al. learn and update a tire model online to account for changes in lateral slip behavior; their evaluation uses numerical and high-fidelity simulators, including simulated changes intended to represent wear [12]. More generally, Brunke et al. identify sparse data and distribution shifts as practical challenges for learning-based control [5]. A change does not necessarily invalidate a learned residual, but performance in one set of conditions does not by itself establish usefulness in another. Chen et al. identify the

limited adaptability of an offline-trained correction to new conditions as a motivation for online updates [7].

The research challenge is to assess during operation whether the residual remains informative for control as the vehicle's response changes, and to determine how that assessment should influence the controller. This requires relating model error to the decisions it affects, without necessarily identifying the physical cause of the discrepancy.

1.2 Research problem

1.3 Scope of the problem (Priority 1)

1.4 Research gap

1.5 Research question

How can the runtime validity of a learned vehicle dynamics residual model be estimated and used to adapt an MPC/NMPC controller under operating conditions not represented during training?

1.6 Specific research questions

RQ1: Which types of physically relevant shifts in vehicle dynamics cause a significant loss of validity in the learned residual?

RQ2: What information do epistemic uncertainty, aleatoric uncertainty, calibration, and an OOD indicator provide about control-relevant model error?

RQ3: To what extent are uncertainty, calibration, and OOD signals complementary or redundant in characterizing runtime validity?

RQ4: How can a confidence/validity signal $C_k = g(U_{\text{epi}}, U_{\text{ale}}, \text{Calibration}, \text{OOD})$ be constructed to reflect the error relevant to the controller?

RQ5: How should C_k be used to adapt the MPC/NMPC horizon to reduce closed-loop performance degradation when the validity of the learned model decreases?

RQ6: What improvement does the proposed architecture offer over a controller based only on the physical model, a controller that uses the learned model without validity monitoring, and architectures that use only validity or only OOD information?

RQ7: As an optional extension, can validity information be used to modify the action of a CBF or trigger data/model update strategies in response to epistemic uncertainty?

1.7 Research hypothesis

A calibrated combination of epistemic uncertainty, aleatoric uncertainty, and evidence of out-of-distribution operation can provide a useful estimate of the runtime validity of a learned vehicle dynamics residual. Incorporating this estimate into the adaptation of an MPC/NMPC controller can reduce closed-loop performance degradation under regime changes compared with architectures that use the learned model without validity awareness.

Chapter 2

Objectives

2.1 Overall objective

Develop and evaluate a methodology for real-time validity estimation for a hybrid vehicle dynamics model comprising a physical model and a learned residual, and integrate this estimate into an MPC/NMPC controller to reduce closed-loop performance degradation under operating conditions not represented during training.

2.2 Specific objectives

SO1: Formulate a hybrid vehicle dynamics architecture comprising a control-oriented physical model and a learned residual.

SO2: Provide operational definitions of the types of shift/OOD relevant to the study, distinguishing at least unrepresented physical regimes and dynamics OOD, and including input OOD if time permits.

SO3: Estimate and calibrate the aleatoric and epistemic uncertainty of the learned model using a heteroscedastic model and a small ensemble, complemented by a computationally inexpensive OOD indicator.

SO4: Experimentally characterize the relationships among distribution shift, uncertainty, calibration, OOD, prediction error, and control-relevant error.

SO5: Define a representation of runtime confidence/validity

$$C_k = g(U_{\text{epi}}, U_{\text{ale}}, \text{Calibration}, \text{OOD})$$

and evaluate its ability to anticipate or represent model degradation in a way that is useful to the controller.

SO6: Integrate the validity signal into an MPC/NMPC controller through horizon adaptation and evaluate its effect on closed-loop performance and empirical robustness.

SO7: Compare the proposed approach with baseline architectures that isolate the effects of the learned model, validity estimation, and OOD detection.

SO8: As an extension, if the schedule permits, evaluate the use of the validity signal in a CBF and/or a DAgger-based data acquisition and retraining strategy for conditions associated with epistemic uncertainty.

Chapter 3

Fundamentals of vehicle dynamics and control

3.1 Vehicle dynamics relevant to control

3.2 Control-oriented models and model mismatch

3.3 Hybrid models: physics and learned residuals

3.4 Model predictive control

3.5 Control barrier functions as an optional extension

Chapter 4

Learned information and runtime validity

4.1 Learned information relevant to the controller

4.2 Predictive uncertainty

4.2.1 Aleatoric uncertainty

4.2.2 Epistemic uncertainty

4.2.3 Calibration

4.3 OOD and distribution shift in vehicle dynamics

4.3.1 Unrepresented physical regime

4.3.2 Dynamics OOD

4.3.3 Input OOD

4.3.4 Sensor failure or degradation

4.4 Runtime validity and confidence

$$C_k = g(U_{\mathrm{epi},k}, U_{\mathrm{ale},k}, \mathrm{Cal}_k, S_{\mathrm{OOD},k}) \,, \tag{4.1}$$

4.5 Control-relevant validity

Chapter 5

State of the art

5.1 Literature review strategy

5.2 Learned and hybrid models for vehicle dynamics control

5.3 Uncertainty-aware learned dynamics for control

5.4 OOD, distribution shift, and runtime monitoring

5.5 Validity-aware and confidence-aware control

5.6 Critical synthesis and research gap

Chapter 6

Conceptual framework and proposed architecture

6.1 The shift–validity–control chain

6.2 Hybrid prediction model

6.3 Validity estimator

6.4 Integration with MPC/NMPC

6.5 Optional extensions

6.5.1 Modulation of a CBF

6.5.2 Data acquisition and retraining

Chapter 7

Research and validation methodology (Priority 1)

7.1 Overall strategy

S1: Establish the control-oriented physical model and the baseline MPC/NMPC controller.

S2: Train the learned residual and quantify its improvement and limitations relative to the physical model within the nominal domain.

S3: Implement aleatoric and epistemic uncertainty estimation, calibration, and a computationally inexpensive OOD score.

S4: Design physically interpretable shifts in vehicle dynamics and characterize their effects on prediction, uncertainty, OOD, and validity.

S5: Construct and validate the signal C_k against control-relevant error.

S6: Integrate C_k into MPC/NMPC adaptation.

S7: Compare the baselines and the proposed approach under nominal and OOD conditions.

S8: Evaluate empirical robustness and, if progress permits, incorporate CBF and/or DAgger extensions.

7.2 Distribution shift scenarios

7.2.1 Unknown change in friction

7.2.2 Extension of the speed range

7.2.3 Sensor failure or degradation

7.2.4 Input OOD

7.3 Learned model and uncertainty

7.4 Construction and evaluation of the validity signal

7.5 Validity-aware integration with MPC/NMPC

7.6 Baselines and ablation study

B0 – Physics-only: Controller using a physical model without a learned residual.

B1 – Learned/no-awareness: Controller using a physical model and a learned residual, without validity estimation or OOD detection.

B2 – Validity/uncertainty-only: Controller using a learned model and a signal constructed from uncertainty/calibration, excluding an explicit OOD component.

B3 – OOD-only: Controller using a learned model and a response based solely on the OOD score, without probabilistic uncertainty estimation.

P – Proposed: Controller using a physical model, a learned residual, and the signal $C_k = g(U_{\text{epi}}, U_{\text{ale}}, \text{Cal}, OOD)$, used to adapt the MPC/NMPC controller.

7.7 Evaluation metrics

7.7.1 Model metrics

7.7.2 Uncertainty, calibration, and OOD metrics

7.7.3 Closed-loop metrics

7.8 Empirical robustness criterion

$$\Delta J_{\text{prop}} < \Delta J_{\text{baseline}}, \tag{7.1}$$

7.9 Level of assurance and scope of the conclusions

7.10 Statistical analysis and reproducibility

Chapter 8

Preliminary results

8.1 Feasibility of learning vehicle dynamics models

8.2 Uncertainty and OOD behavior under distribution shifts

8.3 Preliminary integration with control

8.4 Discussion of preliminary results

Chapter 9

Expected contributions

9.1 Control-relevant validity estimation

9.2 Validity-aware vehicle control

9.3 Runtime validity characterization

9.4 Experimental taxonomy of shifts in vehicle dynamics

9.5 Potential extensions

Chapter 10

Work plan (Priority 1)

10.1 Remaining activities

10.2 Schedule

10.3 Decision milestones

Chapter 11

Risks, limitations, and mitigation strategies

11.1 Risk of redundancy between uncertainty and OOD

11.2 Risk of poor calibration or weak correlation with error

11.3 Risk that horizon adaptation does not improve closed-loop performance

11.4 Computational risk

11.5 Risk of scope expansion

11.6 Limitations of formal guarantees

Chapter 12

Conclusions of the qualifying examination

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